

Path Integrals (in QFT)

Source: M. D. Schwartz, Quantum Field Theory and the Standard Model, Chapter 14.

Path Integral

The intuition for the path integral comes from a simple thought experiment you can do in quantum mechanics. Recall the double-slit experiment: the amplitude for a field to propagate from a source through a screen with two slits to a detector is the sum of the amplitudes to propagate through each slit separately. We add up the two amplitudes and then square to get the probability. If instead we had three slits and three screens, the amplitude would come from the sum of all possible paths through the slits and screens. And so on, for four slits, five slits, etc. Taking the continuum limit, we can keep slitting until the screen is gone. The result is that the final amplitude is the sum of all possible different paths. That is all the path integral is calculating.